

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)**

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No- 11	Team No- 09	Project Title- Car Rental & Mobility Mobile App for Renter Self-Service
-------------------	--------------------	--

1. Abstract:

The **Car Rental & Mobility Mobile App for Renter Self-Service** is a mobile-based application designed to simplify and automate the car rental process by allowing users to independently search, reserve, and manage rental vehicles. The system addresses common problems in traditional car rental services such as manual booking, limited availability information, and dependency on rental staff. Users can register and securely log in, search for available cars based on location and dates, view vehicle details and rental prices, make bookings, complete simulated payments, and track their booking history through a self-service interface. The proposed system uses a **distributed backend architecture** in which separate services handle authentication, vehicle management, booking, payment, and notifications. An **API Gateway** provides a single communication point between the mobile application and backend services. **PostgreSQL** is used for structured data storage, while **REST APIs, JWT authentication, and Docker containers** support secure communication, service isolation, and deployment. The system demonstrates important concepts of **database systems engineering and distributed backend development**, including database design, microservices, API communication, and transaction management. The project aims to provide a scalable, user-friendly, and efficient car rental solution suitable for modern self-service mobility services.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520090063	R. RADHAKRISHNA	
2520090030	R. JAGADEESH	
2520090060	P. VEDADRI NAIDU	

3. Frameworks / Platforms / Software Used:

Frontend: HTML, CSS, JavaScript

Backend: Node.js, Express.js, REST APIs

Database & Tools: PostgreSQL, Docker, VS Code, Postman, JWT Authentication

Guide Name and Signature (remarks if any) _____